



Data-driven modeling for long-term electricity price forecasting

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ABSTRACT

Estimating the financial viability of renewable energy investments requires the availability of long-term, finely-resolved electricity prices over the investment lifespan. This entails, however, two major challenges: (i) the combination of extensive time horizons and fine time resolutions, and (ii) the prediction of out-of-sample electricity prices in future energy and market scenarios, or shifts in pricing regime, that were not observed in the past. This paper tackles such challenges by proposing a data-driven model for the long-term prediction of electricity market prices that is based on Fourier analysis. The electricity price is decomposed into components leading to its base evolution, which are described through the amplitudes of the main frequencies of the Fourier series, and components leading to high price volatility, which are described by the residual frequencies. The former are predicted via a regression model that uses as input annual values of relevant energy and market quantities, such as electricity generation, prices and demands. The proposed method shows capable of (i) predicting the most relevant dynamics of the electricity price; (ii) generalization by capturing the market mechanisms of previously unseen electricity markets. These findings support the relevance and validity of data-driven, finely-resolved, long-term predictions and highlight the potential for hybrid data-driven and market-based models.

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1. Introduction

1.1. Context: Changing electricity markets and deployment of data-driven models

Climate change and global warming have forced the electricity sector into a transition of unprecedented speed [1,2]. Renewable energy sources (RES) are adopted globally to reduce the carbon intensity of electricity generation [3,4]. With the increasing penetration of RES, the supporting policy schemes such as feed-in tariffs (FITs) are being phased out in many markets, leaving RES investors and operators exposed to market prices and investment risks. As investments in new generation assets are capital-intensive and marked by long amortization periods, the estimation of future cash flows of a project is key for supporting investment decisions. Whereas FITs allow for a relatively straightforward estimation of future revenues by only focusing on the estimation of future generation, their phase-out makes it necessary to take the development of future market prices into account. The emergence of

financing tools for RES, such as Power Purchase Agreements (PPAs), further increases demand for electricity price forecasts (EPF), as more non-generating players with scarce expertise in this area are entering the market to contract with RES generators [5,6]. Hence, both operators of RES plants and corporates looking to directly procure renewable energy need to quantify future market prices for evaluating potential revenues and risks. Such forecasting is generally done for different horizons: (i) short-term EPF, in the order of hours or days, and (ii) medium-to long-term EPF, in the order of months or years [7].

Short-term EPF is the field in which data-driven approaches are most commonly applied [8], mostly for electricity trading and short-term operation scheduling [7,9]. In this domain, artificial neural networks (ANNs) have been subject of increased attention during the past decade [10–15], as well as hybrid models coupling ANNs and wavelet transform (WT) [16–19]. Besides these, traditional approaches to short-term EPF have been methods from the domain of time series forecasting, namely autoregressive models such as seasonal and non-seasonal autoregressive integrated moving average (SARIMA/ARIMA) models [20–23], as well as quantile regressions [24].

With an abundance of different statistical models applied to different market prices, price forecasts reported in short-term EPF vary significantly. The approaches that are predominantly applied

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have forecasting horizons of at most a couple of days, rely on hourly resolved input data and on hourly forecasts of exogenous variables, and report mean absolute percentage errors (MAPE) within the range of 10–20%. The difficulty of accurately predicting hourly electricity prices is mainly due to their high volatility [25]. To cope with this challenge, denoising methods like discrete wavelet transform (DWT) [22,26–29], Fourier analysis [30,31] and empirical mode decomposition [32,33] have been applied. This allows to decompose the time series of prices into a combination of periodic terms, which are then predicted separately, or to reduce the volatility of the time series prior to making hourly point forecasts. A comprehensive review of data-driven approaches for electricity price forecasts is provided in Ref. [34].

Medium- and long-term EPF consider forecast horizons of up to several months (medium-term) or years (long-term), and are mostly obtained using econometric, market-based models, which capture the market dynamics by modelling the marginal cost of electricity generators [35–41]. This is due to the ability of market-based models to describe, and potentially anticipate, long-term variations in the relationship between the electricity price and relevant quantities (e.g. policy-driven variations not observed in the past). For this reason, compared to short-term EPF, data-driven medium- and long-term EPF are less explored [8]. Support vector machines have shown capable of modeling monthly-averaged electricity prices in the Nordic and Ontario electricity markets with forecasting horizons of up to one year [42]. Due to the high volatility of electricity prices, however, hourly point-forecasts often lack accuracy when considering long time horizons, for which probabilistic forecasting methods have been adopted [43,44]. Vector error correction models could forecast hourly day-ahead prices with a horizon of one year, reporting MAPEs within the range of 4.6–5.7% for 2011 and 2012 prices in the Spanish market [44]. However, such models incurred in MAPE of 22.5% for 2013 prices, due to unprecedented price decrements in early 2013.

Market-based models circumvent these issues by anticipating the effect of the changing market environment via modelling of the underlying market mechanisms. In other words, they try to predict regime shifts in the electricity market by looking at the underlying physical mechanisms governing the relationship between the electricity price and the connected quantities (e.g. describing the transition from a pricing regime driven by fossil fuel prices to one driven by carbon prices). Long-term predictions of annual mean price obtained with market-based models are made available by a range of governmental agencies, such as the European Commission for several European countries [45], and the British Department of Business, Energy & Industrial Strategy (BEIS) for Great Britain [46]. However, such market-based model predictions are rarely available for finely-resolved (e.g. hourly) electricity prices (examples of models providing hourly-resolved electricity prices are the METIS model adopted by the European Commission [47] and the Nexus-e model developed at ETH Zurich with a focus on Switzerland [40,41]). This is owed to the lack of available hourly-resolved input data (such as electricity demand and generation) for the entire electricity sector and time horizon. Moreover, market-based models require expert judgement that is not guaranteed to be readily available and easily transferable across domains or countries. This is alongside the practical complexity of calibrating market-based models on past and current observations, while aiming to reliably capture future market mechanisms. This leaves them subject to similar drawbacks as data-driven models, which are trained on past and current observations while being often simpler and more easily reproducible.

1.2. Motivation and scope: Challenges of data-driven price predictions

Finely-resolved predictions of electricity market prices are especially important in electricity sectors with high non-dispatchable RES penetration, as the combination of electricity price and renewable generation on an hourly basis can significantly impact the profitability of renewable investments [8,48]. However, determining data-driven correlations between hourly values of future energy scenarios and electricity price comes with two main hurdles. First, information on the future evolution of price drivers is often only available as annual estimates, with hourly forecasts being mainly produced for short-term applications [7,8]. Second, even when the price drivers are available with high resolution, the analysis of historical correlations shows that they exhibit little common patterns with the electricity price due to their high volatility (see for example Fig. A.1 in Appendix A, which shows the historical correlation between electricity price and natural gas price and wind generation from 2015 to 2019).

We aim to develop a data-driven model for the finely-resolved, long-term prediction of the wholesale electricity market price, hereafter denoted as *electricity price*, based on relevant energy and market quantities, denoted as *price drivers*. We do this by discussing and tackling the most relevant shortcomings, namely:

- The combination of long-term time horizons with highly time-resolved predictions to describe the relevant market dynamics, from hourly to yearly to multi-year.

This is addressed by adopting a prediction approach based on Fourier analysis, where the electricity price is described through the sum of trigonometric functions with different amplitudes and frequencies. More specifically, the hourly-resolved electricity price over a year time horizon is decomposed into two terms: (i) its base evolution, which is described through the amplitudes of the dominant frequencies and is responsible for the main price behavior, and (ii) its volatility, which is described by all the residual frequencies and is responsible for extreme price values. The main idea behind our approach is that the base evolution of the electricity price can still be determined by looking at the average annual values of the price drivers (see Fig. A.2 in Appendix A, which reports the correlation between the average annual values of price drivers and relevant fluctuations of the electricity price). The dominant frequencies are defined based on the historical behavior of electricity prices and are, for example, the annual, weekly, daily, or intra-daily fluctuations. The amplitude of each frequency corresponds to one value defining the corresponding price dynamics for the underlying year. These amplitudes are the quantities predicted by the data-driven model based on the annual values of price drivers. The volatility of the electricity price, as well as its extreme values, are accounted for by adding a random residual profile (i.e. residual frequencies) obtained from historical hourly-resolved time series.

A similar approach was proposed by Yukseltan and co-workers to forecast electricity demand over a one year time horizon [31]. We build on this to forecast electricity price over a multi-year time horizon. Our approach does not aim at providing point forecasts of hourly market prices for electricity trading or operation optimization purposes, but rather realistic hourly-resolved price profiles for the valuation of existing or planned generation assets.

- The rapid transformations of the electricity sector and the unprecedented variety and variability of exogenous quantities

affecting the electricity price, which result in shifts in pricing regime, hence in out-of-sample issues. Such out-of-sample issues mostly arise because of (i) scarcity of data, as a consistent set of historical values of electricity price and corresponding price drivers is often available only for a few countries and for recent years; (ii) unprecedented, oftentimes policy-driven, changes in the evolution of the price drivers and of their underlying correlations with the electricity price. These issues are tackled by coupling the proposed data-driven model with predictions of average annual electricity prices obtained via market-based models.

- The uncertainty associated with the model input data, i.e. the price drivers. Inspired by Ref. [49], the uncertainty associated with an price drivers is characterized via the empirical prediction interval approach. The uncertain price drivers are used within a Monte Carlo simulation to characterize the uncertainty associated with the price prediction.

Whereas the proposed method is general and can be applied to any electricity market, here it is presented on the case-study of the United Kingdom (UK). Electricity market prices from the bidding zone of Great Britain are considered for the UK. Furthermore, data from Sweden (SE), Germany (DE) and Denmark (DK) are also used for assessing model performance. More specifically, the capability of the model to make predictions for a previously unseen electricity market is evaluated through a cross-country assessment.

The paper is structured as follows. Section 2 describes the methods and tools used to perform the analysis. Section 3 presents and discusses the results of the analysis, and contrasts the proposed solution against state-of-the-art approaches. Finally, Section 4 draws conclusions.

2. Model description

The developed data-driven model relies on (i) the selection of the input data used to predict the electricity market price, (ii) the description of the price profile through its most relevant features, (iii) the description of price volatility and extreme values, (iv) the selection of regression models, (v) the coupling of data-driven and market-based models, and (vi) the assessment of the uncertainty associated with price predictions.

2.1. Selection of the input data

Table 1 lists the set of all price drivers taken into consideration as inputs to the data-driven model. This set reflects the inputs typically used within fundamental, market-based models. It includes electricity generation and demand values, as well as fossil fuels and carbon prices, and aims to reflect the most relevant factors defining the electricity price. For all price drivers, two types of input data are collected: (i) historical values, which are used to train and validate the model, and (ii) future predictions, which are used to determine future values of the electricity price. Historical values of price drivers and electricity price are available from 1990 to 2019 with annual resolution, whereas historical hourly-resolved values are available from 2015 to 2019.

Most historical data is collected from Eurostat [50], with the exception of fossil fuels and carbon prices, which are collected from the latest British Petroleum (BP) report [51]. Concerning the future evolution of the price drivers in the UK, several scenarios are provided by sources such as BEIS [46] and National Grid [52] in the UK and by the European Commission [45] for several European countries. Such scenarios do not always cover all the price drivers and can span different time horizons. Here, a scenario is defined that consists of data from the reference scenarios of BEIS and of the European Commission econometric models (as detailed in Table 1), covering the time horizon 2020–2035. This allows for a consistent set of input data, which is denoted as the *reference scenario* hereafter.

When considering all available price drivers, interdependence issues might arise. Furthermore, with different technologies contributing to the overall electricity generation, a driver relevance is expected to vary when considering different markets. We address these points via a sensitivity study that investigates the possibility of a reduced set of price drivers improving the prediction performance.

2.2. Fourier analysis of electricity price profile

While the analysis of historical data shows that the price drivers exhibit no significant correlation with the electricity price on an hourly scale, the main idea behind the proposed approach is that a correlation exists between the base evolution of the electricity

Table 1

Price drivers used as input data to the price prediction model. Reference scenarios of BEIS and European Commission used for future predictions. A fixed conversion factor of 1.1 EUR/GBP is adopted.

Price driver	Unit	Source, Historical values	Source, Future predictions
<i>Total electricity sector</i>			
Electricity demand	GWh	Eurostat [50]	European Commission [45]
Electricity generation	GWh	Eurostat [50]	European Commission [45]
Electricity imports	GWh	Eurostat [50]	European Commission [45]
<i>Electricity generation mix</i>			
Wind generation	GWh	Eurostat [50]	European Commission [45]
Solar generation	GWh	Eurostat [50]	European Commission [45]
Biomass generation	GWh	Eurostat [50]	European Commission [45]
Hydro generation	GWh	Eurostat [50]	European Commission [45]
Geothermal generation	GWh	Eurostat [50]	European Commission [45]
Nuclear generation	GWh	Eurostat [50]	European Commission [45]
Natural gas generation	GWh	Eurostat [50]	European Commission [45]
Oil generation	GWh	Eurostat [50]	European Commission [45]
Coal generation	GWh	Eurostat [50]	European Commission [45]
<i>Fossil fuel prices</i>			
Natural gas price	EUR/MWh	BP [51]	BEIS [46]
Oil price	EUR/MWh	BP [51]	BEIS [46]
Coal price	EUR/MWh	BP [51]	BEIS [46]
<i>Policy instruments</i>			
Carbon price	EUR/tCO ₂	BEIS [46]	BEIS [46]

price and the average annual values of the price drivers. The base evolution of the electricity price corresponds to its most relevant fluctuations, such as annual, weekly, daily, or intra-day fluctuations. To illustrate, consider the so-called price cannibalization effect, namely the depressive influence on the electricity price at times of high output from intermittent renewable generation. For example, high shares of solar penetration, which can be described by high values of annual solar generation, result into low values of the electricity price in summer and during the day, and into a greater seasonal and intra-day variability of the electricity price [53].

In other words, a correlation exists between annual price drivers and the base evolution of the electricity price, which can be decomposed into a limited number of spectral components. Following this idea, the information on the annual average values of the price drivers can be used to predict future trends of the hourly price profile. We apply Fourier analysis to the hourly-resolved profiles of electricity price over an annual time horizon. Fourier analysis allows to represent or approximate any generic function through a Fourier series, i.e., the sum of trigonometric functions with different frequencies and amplitudes [54]. In the case of the electricity price profile, p_t , the Fourier series representation is given by Eq. (1):

$$p_t = a_0 + \frac{2}{N} \sum_{n=1}^{N/2} a_n \cos\left(\frac{2n\pi t}{N}\right) + b_n \sin\left(\frac{2n\pi t}{N}\right) \quad (1)$$

where a_0 corresponds to the mean value of p_t , a_n and b_n are the Fourier coefficients defining the shape of the periodic functions at frequency n , and N is the length of the original time series. The subscript t refers to the time instant at which the Fourier series is evaluated to obtain the values of the original time series. These coefficients are determined via discrete Fourier transform (DFT) [55]. As we apply DFT on annual time series with hourly resolution, $N = 8760$ and $t \in \{1, \dots, H\}$ with $H = 8760$ is the total number of hours in a year. In this case, a_0 represents the mean annual value of the electricity price. The Euclidean norm of parameters a_n and b_n defines the amplitude of the periodic component with frequency n . The complete time series is described by a set of N coefficients, which are constant on the entire domain of the sequence, i.e. on one year. To obtain the value of the original time series, the discrete Fourier series is evaluated at time instant t . Therefore, the entire annual time series is described through a set of annual coefficients.

To predict the base evolution of the electricity price profile, only the frequencies of the Fourier series with the largest amplitude are considered. These are denoted as *main frequencies*, n_{main} . Thereby, a smoothed profile capturing only the major dynamics of the electricity price is obtained. We refer to this as *price approximation*, which is defined as

$$f_t = a_0 + \frac{2}{N} \sum_{n \in n_{\text{main}}} a_n \cos\left(\frac{2n\pi t}{N}\right) + b_n \sin\left(\frac{2n\pi t}{N}\right) \quad (2)$$

where f_t is the approximate electricity price at time instant t , which consists of the main frequency components of the Fourier series.

While the price approximation captures the base evolution of the electricity price, hence the most relevant price dynamics, it is unable to describe hourly fluctuations or times of extreme prices that go beyond the base patterns of the price profile. These are described by a residual term, R_t , which consists of all the residual frequencies not included into the price approximation. Thus, the electricity price can be expressed as:

$$p_t = f_t + R_t \quad (3)$$

Overall the proposed approach is based on the following steps:

- (i) the Fourier coefficients describing the hourly profile of historical electricity price are determined; (ii) regression models are trained by using price drivers to predict the Fourier coefficients corresponding to the price approximations (see Section 2.3); (iii) future values of the Fourier coefficients of the price approximation are predicted based on future values of the price drivers; (iv) future values of the Fourier coefficients of the residual terms are obtained by sampling historical residual terms.

2.3. Model selection and performance assessment

The price approximation is based on the decomposition into the frequency domain and predicting the associated coefficients using a regression model. The model takes as an input the annual values of the price drivers and produces as outputs the Fourier coefficients of the price approximation. A comparative assessment of different regression models is carried out, which includes linear regression, Gaussian process regression (GPR) [56,57] and artificial neural networks (ANN) [10,58]. For all models, two approaches are considered:

- Single-output. The Fourier coefficients describing the annual price approximation, i.e. a_n and b_n with $n \in n_{\text{main}}$, are predicted using separate models. The advantage of this approach stems from the potentially different relationships between different coefficients and price drivers, where all these relationship can be tuned individually.
- Multi-output. The Fourier coefficients describing the annual price approximation are predicted using a single model. The advantage of this approach derives from the possibility of taking into account the potential interdependence of the predicted coefficients.

In the case of linear regression, various regressor configurations are investigated, namely linear, quadratic, and logarithmic relationships; in the case of GPR, different kernel configurations are compared; concerning ANN, the number of neurons and layers is varied and different activation functions are compared. All models are trained using historical annual values of the price drivers and historical values of the Fourier coefficients (computed from the historical hourly-resolved time series of electricity price). Model training, validation, and selection is performed on the UK reference scenario. Each model is tested via leave-one-out cross-validation, where the training set consists of four years and the validation set consists of the remaining year. The MAPE of the price approximation function is used to assess the prediction performance of the different models. It is defined as:

$$\text{MAPE} = \frac{100}{H} \sum_{t=1}^H \frac{|\hat{f}_t - f_t|}{|f_t|} \quad (4)$$

where $\hat{f}_t$ is the t -th prediction of function f and f_t its observed value.

First, the optimal model, i.e. the one leading to the minimum MAPE, is determined and assessed with reference to the UK case-study. Then, the capability of the model to make predictions for a previously unseen electricity market is evaluated through a cross-country assessment, where the optimal model is (i) determined by using data from UK, (ii) trained by using data from UK, Germany and Sweden, and (iii) validated on Denmark (Danish bidding zone 1). This allows (i) to test whether the methodology developed can be used on a market where historical data is not readily available for training, and (ii) to understand whether data from different markets can be used to increase the available training set.

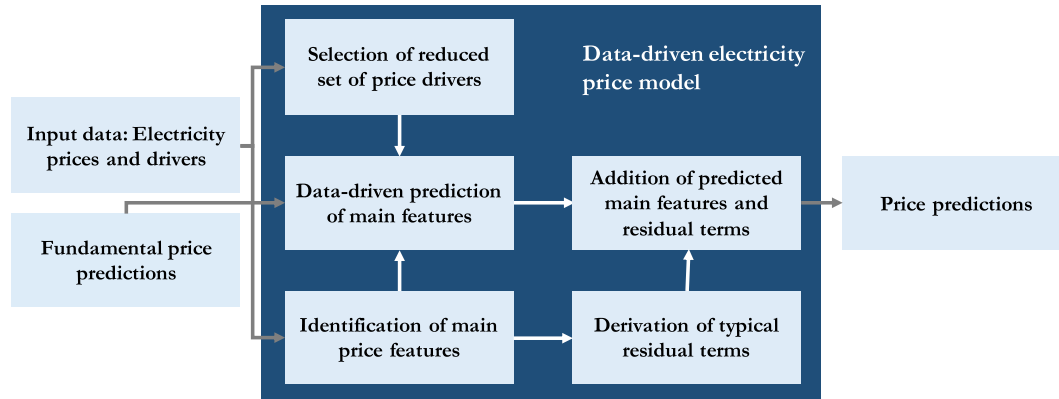


Fig. 1. Overview of the structure of the data-driven model.

2.4. Coupling data-driven and market-based models

A major challenge arising when using regression models to predict future electricity prices is given by out-of-sample issues, which consist in testing the model on data that is unseen during training and validation. This is mostly due to scarcity of historical data and to unexpected, oftentimes policy-driven, evolution of the price drivers. A good example of this is given by the carbon price: whereas low carbon prices are observed historically (i.e. within the 2015–2019 training interval), most scenarios project carbon prices to increase significantly in the future (i.e. during the 2020–2035 the testing time horizon). This implies that the relationship between the carbon and the electricity price will most likely change in the future, though this cannot be captured during the model training phase.

We tackle out-of-sample issues through two distinct approaches.

- Coupling of data-driven and market-based models. Market-based models provide information about possible future correlations between price drivers and electricity price. This information is often available only in terms of mean annual electricity price. Nevertheless, the data-driven model can account for it by including future annual values of price drivers and electricity price into the training data. This allows the data-driven model to learn from the first-principle process predictions obtained from the market-based models.
- Introducing artificial damping mechanisms. The impact of a given price driver on an electricity price feature can be artificially modified by introducing a fictitious variable that offsets its effect. The damping variable mirrors the behavior of the corresponding price drivers within the training data and evolves inversely within the testing data (i.e. the data used for prediction). In this way, the damping variable prevents extreme price values due to the out-of-sample variation of a price driver. From a physical standpoint, a damping variable can be interpreted as the market reacting to drastic changes in a price driver and adapting to the new circumstances and external factors, hence driving against the impact of extreme price driver developments.

2.5. Uncertainty assessment

The uncertainty associated with the future evolution of price drivers is characterized by adopting an empirical prediction interval approach [49], and by projecting the past forecast errors committed by the adopted UK reference scenario. Forecast errors

are assumed to be normally distributed, with zero mean (method G1 presented in Ref. [49]) and with standard deviation increasing linearly with time. Such errors are used to build uncertainty distributions around the point forecasts of the corresponding price drivers (i.e. the reference scenario defined in Section 2.1).

The resulting probability distribution of electricity price is obtained via Monte Carlo simulation, where the forecast errors are sampled from a multivariate normal distribution. The covariance matrix of the distribution is defined based on historical correlations and standard deviations of price drivers from 2012 to 2018:

$$\text{cov}(X, Y) = \frac{\rho_{XY}}{\sigma_X \sigma_Y} \quad (5)$$

where ρ_{XY} is the Pearson correlation coefficient of variables X and Y , σ_X and σ_Y are the standard deviations of X and Y , respectively, and $\text{cov}(X, Y)$ their covariance.

2.6. Model summary

The data-driven price model is summarized in Fig. 1 and comprises the following steps:

- (1) Fourier analysis is applied to determine the coefficients of the Fourier series describing the hourly profiles of historical electricity price.
- (2) Regression models are trained by using historical annual values of price drivers as input and historical Fourier coefficients of the price approximation (i.e. base evolution of the electricity price) as output. Here, the data-driven model can be coupled with market-based models by using their price prediction during the training phase.
- (3) Regression models are validated on available historical data, which are not used for training. The optimal regression model and the optimal combination of price drivers are obtained.
- (4) Future values of the Fourier coefficients of the price approximation are predicted starting from future values of the price drivers.
- (5) The predicted price approximation is superimposed with a randomly-sampled residual term.
- (6) Steps (4)–(5) are repeated according to a Monte Carlo simulation for several scenarios of price drivers to characterize the uncertainty associated with the price prediction.

3. Results and discussion

This section describes the results of the data-driven predictive

model when applied to the UK case-study and discusses the major advantages and shortcomings of the proposed method.

3.1. Price driver selection

To investigate the relevance of the price drivers and to investigate possible interdependence among them, the following sensitivity study is carried out. First, the input data is pre-screened to remove all price drivers that are not relevant for the market under consideration (i.e. generation technologies that are not in use), resulting in a set of M price drivers that are considered. Then, all subsets of m price drivers, with $m \leq M$, are assessed in terms of overall model prediction performance (i.e. minimum MAPE, see Section 2.3).

The model predictive performance as a function of the number of price drivers is shown in Fig. 2, which reports the minimum MAPE obtained when considering all possible combinations of the M price drivers, considered m at a time. It indicates the optimal combination of price drivers. The results for the single-output GPR model are reported, as this leads to the highest model performance (see Section 3.3).

Prior to the analysis, oil generation and geothermal generation are removed from the input set, as they currently do not contribute to UK power generation and are not foreseen to contribute significantly within the 2020–2035 time horizon. Further, significant policy-driven variations have to be accounted for: coal generation is scheduled to phase out by 2024 [59,60]. Therefore, coal generation and coal prices are removed from the input set too. This leaves a total of $M = 12$ price drivers (i.e. 12 possible combinations for $m = 1$, 1 possible combination for $m = 12$).

Three intervals are identified.

- For $m < 2$, reducing the number of price drivers leads to a decrease of the model predictive performance, suggesting underfitting due to too little information being captured with such a small number of price drivers.
- For $2 \leq m \leq 4$, the model predictive performance are essentially independent of m .
- For $m > 4$, increasing the number of price drivers leads to a decrease of model performance, suggesting overfitting due to interdependence and redundancy issues.

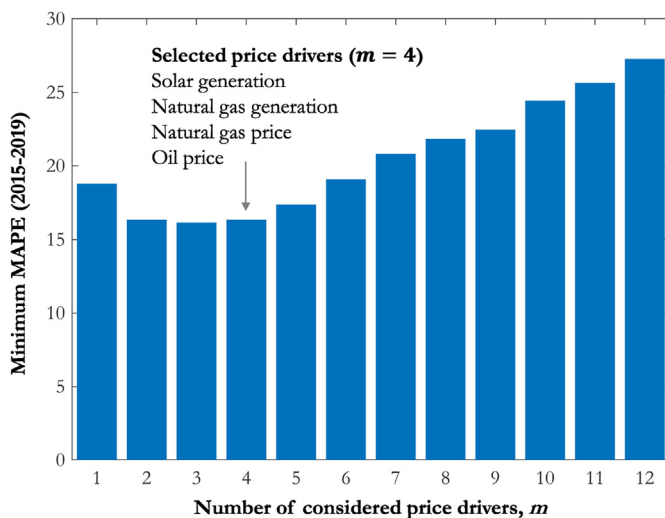


Fig. 2. Minimum mean absolute percentage errors (MAPE) obtained when computing historical values of main frequencies for all possible combinations of $M = 12$ price drivers, considered m at a time ($M!/m!(M - m)!$), with the single-output GPR model.

Based on this, a number of $m = 4$ price drivers is selected as the optimal trade-off leading to a representative model while avoiding overfitting. These price drivers are solar generation, natural gas generation, natural gas price, and oil price.

The same procedure can be generally applied by considering future price scenarios, rather than historical prices, to account for potential variations in the relationship between the price drivers and the electricity price (e.g. to describe a shift from a pricing regime driven by fossil fuel prices to one driven by carbon prices).

3.2. Price feature identification

The main frequencies describing the base evolution of the electricity price are identified as the frequencies of the DFT with greatest amplitude, when considering historical price profiles from 2015 to 2019. More specifically, the six most relevant frequencies are selected, since this allows maintaining a number of features that can be handled by the data-driven model and results in the highest predictive capabilities. Hereafter, we simply refer to the amplitude of the main frequencies, i.e. the Euclidean norm of coefficients a_n and b_n , to provide a single, more interpretable value. For all years in the training set and for all countries of interest, Table 2 lists the amplitude of the main frequencies, $n \in n_{\text{main}}$, and ranks them from the most (left-hand side) to the least relevant (right-hand side). While the annual mean ($n = 0$) is consistently found to be most prominent, the rank of other main components varies depending on country and year.

Concerning UK, the most significant frequencies are $n_{\text{main}} = \{0, 1, 365, 730, 1095, 2\}$, which corresponds to the average, annual, daily, twelve-hourly, eight-hourly, and six-monthly dynamics. Comparing such frequencies with the cases of Germany, Sweden and Denmark, highlights that some of the most relevant dynamics differ across countries. More specifically, the eight-hourly component ($n = 1095$) is less significant in these countries, whereas the weekly dynamics ($n = 52$) is consistently present. This implies that the intra-day variability is less pronounced in Germany, Sweden and Denmark than in the UK, perhaps due to the high fraction of dispatchable generation technologies. As we target a set of price components that is consistent across different years and countries, the weekly component of the Fourier series is also included into n_{main} for the prediction of UK electricity prices. The resulting set of main frequencies is $n_{\text{main}} = \{0, 1, 52, 365, 730, 1095\}$.

Referring to Eq. (3), the two terms of the electricity price given by the main frequencies (price approximation, f_t) and residual frequencies (R_t) are illustrated in Fig. 3 for the case of UK and 2017. The price approximation (red) represents a smoothed version of the actual price profile, only capturing the main price dynamics, whereas the residual terms (gray) describes the price volatility and extreme price values. Fig. 3-a shows hourly resolved prices for an exemplary day, while Fig. 3-b shows daily mean prices over the entire year. Overall, the main price frequencies allow capturing the intra-day patterns given by morning and evening peaks, as well as seasonal patterns given by lower prices in summer and higher prices in winter.

3.3. Model selection and performance assessment

The price approximation coefficients, i.e. a_n and b_n with $n \in n_{\text{main}}$ (see Eq. (2)), are obtained by comparing different regression models, namely linear regression, GPR and ANN, with both single- and multi-output approaches. The comparison is presented in Table 3, which reports the MAPE (see Eq. (4)) corresponding to the optimal tuning for all regression models.

The comparative assessment shows that GPR leads to the highest performance (minimum MAPE) for both single- and multi-

Table 2

Amplitudes of the main frequencies of Fourier series for United Kingdom (UK), Germany (DE), Sweden (SE) and Denmark (DK). Amplitudes are defined as the Euclidean norm of the a_n and b_n coefficients to provide a single, more interpretable value. They are listed in decreasing order from left to right, with the corresponding frequency (n) reported in brackets. The average amplitude of all residual terms is 0.068 EUR/MWh. Discrete Fourier Transform (DFT) implemented through the Python package *numpy* [55].

Country	Year	Amplitude of main frequencies at (n) [EUR/MWh]						
UK	2015	45.52 (0)	3.80 (365)	3.34 (730)	2.36 (1095)	1.48 (1)		1.29 (1460)
	2016	45.66 (0)	6.12 (365)	5.24 (1)	5.20 (730)	4.49 (1095)		3.38 (2)
	2017	51.40 (0)	4.18 (730)	3.93 (1)	3.92 (365)	2.77 (1095)		1.48 (1460)
	2018	65.19 (0)	4.37 (730)	3.88 (365)	3.70 (1)	2.70 (1095)		1.82 (4)
	2019	48.49 (0)	3.83 (1)	3.69 (730)	3.35 (365)	2.06 (2)		1.86 (4)
DE	2015	31.62 (0)	3.46 (730)	2.85 (52)	2.68 (365)	2.07 (3)		1.45 (104)
	2016	28.99 (0)	3.47 (1)	2.64 (730)	2.27 (365)	1.84 (52)		1.63 (2)
	2017	34.20 (0)	3.50 (52)	3.32 (730)	2.94 (365)	2.67 (1)		2.21 (6)
	2018	44.90 (0)	5.86 (1)	3.83 (730)	2.93 (52)	2.90 (365)		1.52 (15)
	2019	37.67 (0)	3.57 (730)	2.80 (52)	2.46 (4)	2.34 (365)		1.90 (30)
SE	2015	22.90 (0)	3.21 (1)	2.02 (365)	1.78 (5)	1.46 (730)		1.38 (3)
	2016	29.55 (0)	3.53 (1)	2.44 (365)	2.08 (2)	1.58 (730)		1.20 (5)
	2017	32.18 (0)	2.30 (365)	1.59 (730)	1.54 (52)	1.38 (1)		0.97 (5)
	2018	46.34 (0)	4.04 (1)	3.06 (365)	2.46 (3)	2.38 (730)		2.04 (4)
	2019	39.80 (0)	3.06 (1)	2.54 (4)	2.35 (365)	2.26 (730)		1.72 (2)
DK	2015	22.89 (0)	2.45 (365)	1.91 (1)	1.82 (730)	1.57 (2)		1.41 (52)
	2016	26.68 (0)	2.98 (1)	1.73 (365)	1.66 (2)	1.51 (730)		1.25 (8)
	2017	30.09 (0)	2.07 (365)	2.05 (730)	1.69 (52)	1.15 (18)		1.10 (25)
	2018	44.05 (0)	4.79 (1)	2.93 (730)	2.80 (365)	2.28 (3)		1.81 (52)
	2019	38.49 (0)	2.92 (730)	2.46 (4)	2.26 (365)	2.24 (52)		1.60 (6)

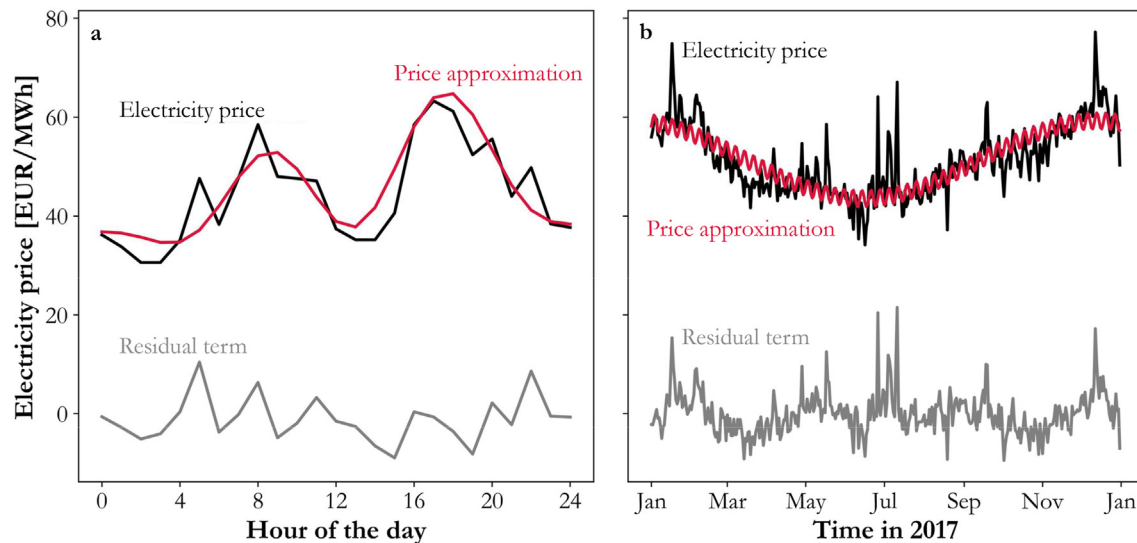


Fig. 3. Two components of electricity price profile (black), namely the price approximation (red) and the residual term (gray). (a) Hourly-resolved prices for an exemplary day, (b) daily mean prices over the year. Electricity price in the UK in 2017.

Table 3

MAPE of all regression models. The performance corresponds to the optimal parametric tuning of the algorithms.

	Linear regression	GPR	ANN
Multi-output	18.9%	13.5%	14.3%
Single-output	19.2%	13.2%	15.2%

output approaches. Whereas both approaches lead to similar values of MAPE, the multi-output approach proves to be numerically less stable, due to the difficulty in defining covariance functions that reliably capture the correlation between the model output variables; this is consistent with other literature studies [61,62]. Therefore, the single-output GPR model is selected as the optimal regression model. For all price approximation coefficients, the optimal kernel functions are reported in Table 4.

The GPR model predictive performance is further investigated.

Fig. 4 reports the predicted electricity price versus the registered historical values in the UK for (a) mean annual electricity prices from 2015 to 2019, and (b) hourly electricity prices during the first week of May for two exemplary years, namely 2016 and 2018. Fig. 4-a shows that the model provides accurate predictions of the mean annual electricity price for 2016 and 2017, for which historical

Table 4

Optimal kernel functions for the single-output GPR model when comparing Linear (Lin), periodic (Per) and radial basis function (RBF) for the UK case-study. The * symbol indicates the product of two functions.

Price approximation coefficient	Kernel function
$a_0, a_{52}, a_{365}, b_{365}, a_{1095}, b_{1095}$	Lin
a_1, b_{730}	RBF + Per
b_1	Lin + RBF + Per + RBF * Per
b_{52}	Lin + RBF * Per
a_{730}	Lin + RBF + Per

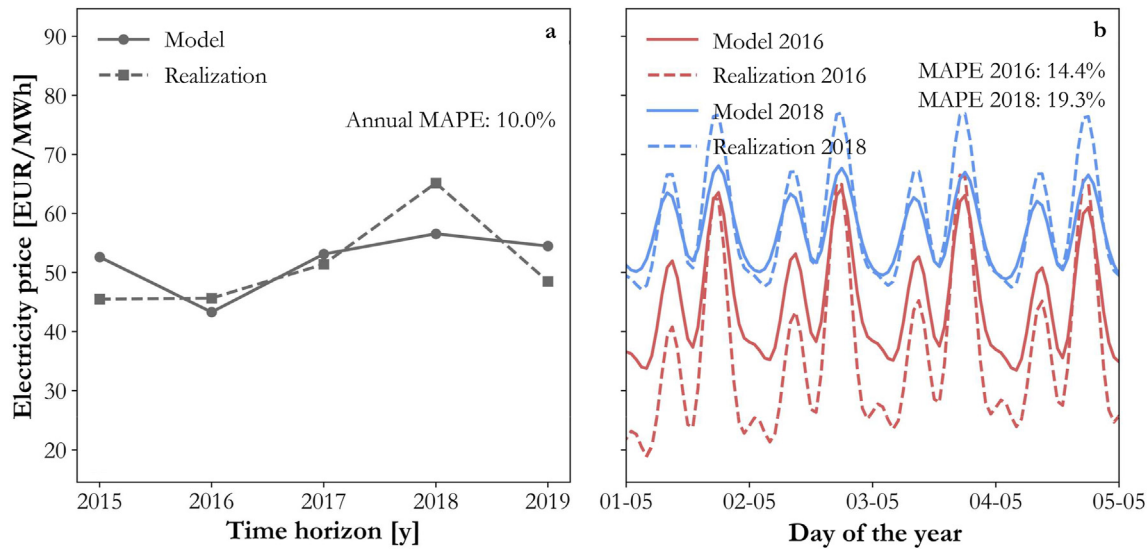


Fig. 4. Predicted electricity price versus the registered historical values in the UK for (a) mean annual electricity prices from 2015 to 2019, and (b) hourly electricity prices during the first week of May in 2016 and 2018.

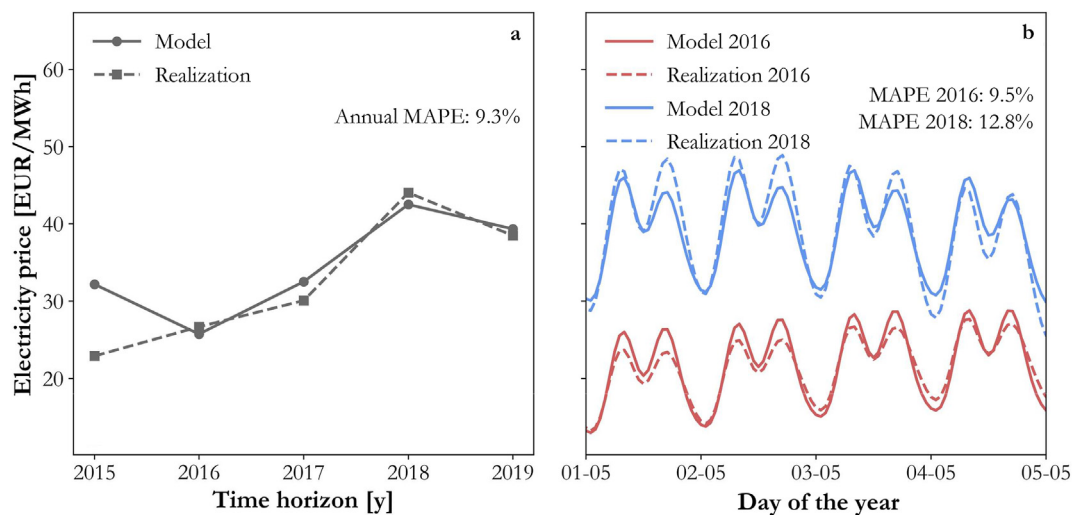


Fig. 5. Predicted electricity price when using UK, DE and SE to train the model and DK to validate it versus the registered historical values for DK for (a) mean annual electricity prices from 2015 to 2019, and (b) hourly electricity prices during the first week of May in 2016 and 2018.

realizations fall within the overall range of training values (MAPE = 5.7% and 3.6% for 2016 and 2017, respectively). Greater MAPEs are observed for 2015, 2018 and 2019 (though always lower than 16%, see Table 5), for which historical realizations are at the boundaries of the range of training values. Over-predictions are observed for 2015 and 2019 and under-prediction is observed for 2018. On an hourly basis, the model captures well the most relevant price dynamics and is able to describe moments of low and high prices across the year (Fig. 4-b). An offset bias might be present in certain periods of the year because of discrepancies in low-frequency coefficients.

In addition to the UK case-study, a cross-country assessment is carried out and shown in Fig. 5. This suggests that both on an annual and on an hourly basis the model provides a good price approximation also on countries different than those used during the training phase. On an annual basis (Fig. 5-a), the overall price trends are captured with the exception of 2015. Similar to UK, this can be explained with predictions being made at the boundaries,

combined with the effect of actual Danish prices in 2015 being lower than 2015 prices in the UK, Germany and Sweden, which the model is trained on.

This reflects an out-of-sample issue, which can be tackled (i) by broadening the set of available training data, which makes more likely to apply the model to already seen values of price drivers and electricity price, and (ii) by including additional information on the validation country during the training phase.

On an hourly basis (Fig. 5-b), the development of the main price dynamics is captured with a comparable accuracy than in the UK case-study (Fig. 4-b), considering only the set of price drivers found to be most relevant for the UK electricity price (simulating the condition where no information is available on the validation country). This suggests data from different markets can well be taken into account. However, the applicability of the price driver set depends on the underlying market conditions and might not always hold true, for example when the energy mix of the countries is very different.

Table 5

Comparison of predictive performance (with corresponding sign, positive/negative when the model overestimates/underestimates realizations) of the proposed data-driven model and the market-based model used by BEIS for predicting the mean annual price in the UK from 2010 to 2019.

Agency report	Mean percentage error [%]									
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
BEIS 2010	26.6	15.5	28.1	18.0	46.0					
BEIS 2014						29.9	32.4	15.9	−12.7	14.4
BEIS 2015						8.30	13.4	11.7	−18.1	5.26
BEIS 2016							−13.5	−18.2	−35.2	−13.0
BEIS 2017								0.41	−22.6	2.97
BEIS 2018									−1.2	25.9
BEIS 2019										9.84
Data-driven model	−4.55	−11.31	−4.85	−11.85	3.74	16.0	−5.73	3.60	−13.3	12.4

Finally, to put the model predictive performance into perspective, the mean percentage error (with corresponding sign) on the annual electricity prices is compared against that committed by the BEIS market-based models. The annual electricity price is considered as this is the only quantity provided by market-based models. Two distinct comparisons are carried out and reported in Table 5:

- For the time horizon 2015–2019, i.e. time horizon used for training the model, against the different editions of the BEIS reference scenario [63]. The predictions provided by the BEIS reports cover future years, and possibly the report year depending on the time of the year when the report is issued. The prediction performance of the proposed data-driven model is comparable to the BEIS market-based model, and is superior in various instances. The wide range of MAPEs committed by BEIS, ranging from 0.41% for 2017 predictions made in 2017 to −35.2% for 2018 predictions made in 2016, stresses the difficulty of accurately predicting electricity prices, even via models describing market-based market mechanisms.
- For the time horizon 2010–2014, which is not used during training phase, against the BEIS reference scenario published in 2010. The comparison is also shown in Fig. 6, and illustrates the performance of the data-driven model over a time horizon not used for training the model. It serves to simulate future time horizons. Also in this case, the data-driven model describes the values and the dynamics of the electricity price better than the BEIS 2010 projection, which is relatively constant over time and constantly overestimates the electricity price.

3.4. Addressing of out-of-sample issues

Out-of-sample issues arise when testing the model on data that are unseen during training and validation, i.e. when overlooking some of the factors leading to variations in the electricity price. When variations are due to policy measures that produce a change in the correlation between the electricity price and the price drivers, two ways of minimizing out-of-sample issues consist of (i) using information coming from market-based models to make implications on the impact of such policy measures, and (ii) artificially reducing their effects through heuristic considerations (e.g. the damping mechanisms described in Section 2.4).

Both approaches are investigated in Fig. 7, which shows the evolution of average annual electricity prices from 2020 to 2035 (Fig. 7-b) as provided by the BEIS 2018 market-based model (diamond, blue solid line), the data-driven model (circle, gray solid line), the data-driven model coupled with the BEIS model (downside triangle, dark blue solid line), and the data-driven model implementing damping effects (upside triangle, green solid line). All these profiles are obtained by considering the same price drivers evolution, i.e. the reference scenario shown in Fig. 7-a. Both the data-driven and the market-based models predict an increase of the mean annual electricity price. However, when correlating the predictions with the price drivers' evolution highlights one major aspect. The effect of fossil fuel prices on the electricity price is significantly higher for the data-driven model than for the market-based model. More specifically, the price of natural gas becomes the most relevant prediction parameter and prevents the other price

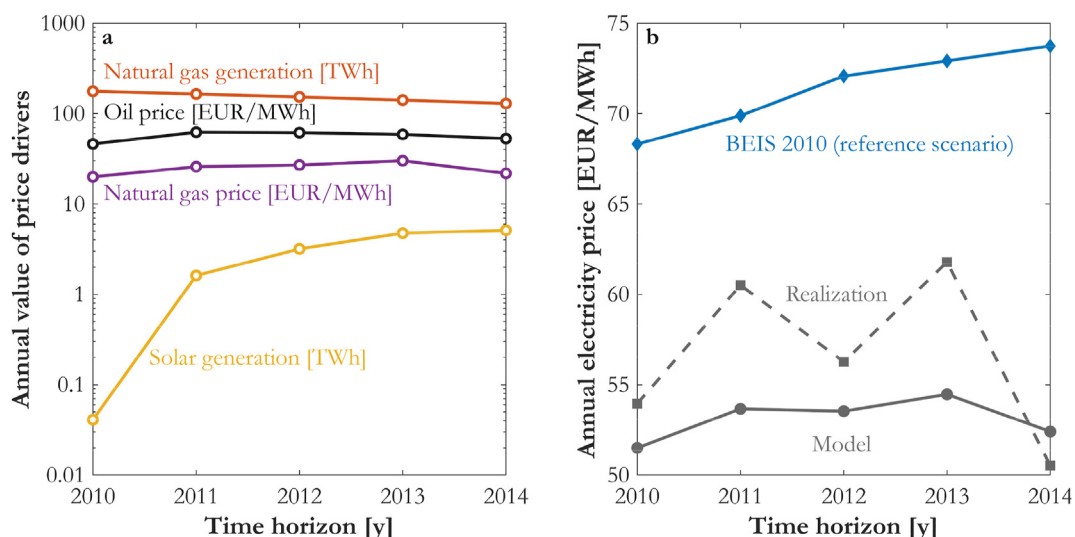


Fig. 6. (a) Price drivers realizations from 2010 to 2014. (b) Average annual electricity price from 2010 to 2014: price realizations (squares, gray dashed line); prediction of BEIS 2010 market-based model (diamond, blue solid line); prediction of data-driven model (circle, gray solid line).

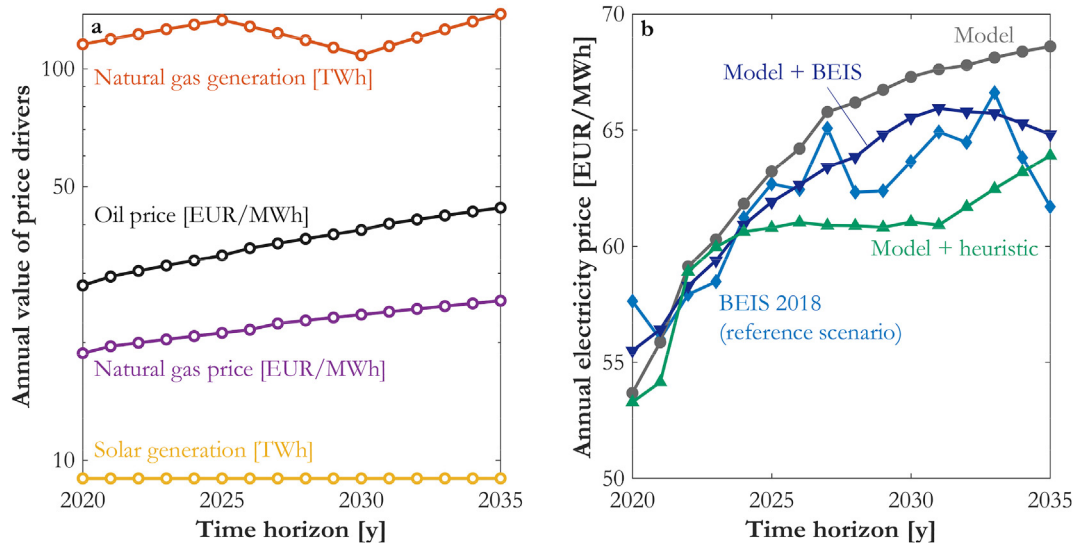


Fig. 7. (a) Evolution of the price drivers corresponding to the UK reference scenarios from 2020 to 2035. (b) Evolution of the average annual electricity prices from 2020 to 2035: prediction of BEIS 2018 market-based model (diamond, blue solid line); prediction of data-driven model (circle, gray solid line); prediction of data-driven model coupled with BEIS 2018 market-based model (downside triangle, dark blue solid line); prediction of data-driven model with damping of out-of-sample price drivers (upside triangle, green solid line).

drivers to influence the dynamics of the electricity price. Comparing the expected evolution of natural gas prices to the evolution of the other price drivers reveals that this might be caused by the fact that the model is trained on values of natural gas price lower than those used for the prediction. This results in a steeper increase of the electricity price for the data-driven model than for the market-based model.

Assuming that a market-based model can better capture the interplay between natural gas and electricity prices, we address this by coupling the data-driven model with market-based predictions. In this way, the information available through the latter allows the former to revise the predicted impact of natural gas price on the electricity price, and to better balance its impact on the price dynamics. Similarly, introducing an artificial price driver damping the effect of natural gas price allows to offset the corresponding variation of electricity price.

The relative offsetting effect depends on how the damping price driver is defined, and is often determined following market-based considerations. While neither approach guarantees better predictions of the electricity price, they both allow to account for a wide set of considerations that might be difficult to take into account when using data-driven models.

Overall, coupling data-driven and market-based models results into a more straightforward implementation than introducing a damping effect, which requires an in-depth analysis of data-driven predictions and the definition of an additional variable based on market-based considerations. Therefore, model coupling is considered as the preferred strategy for those countries where predictions based on market-based models are readily available.

3.5. Uncertainty of price prediction

The uncertainty associated with future electricity price predictions is quantified by coupling the prediction model and Monte Carlo simulation after characterising the uncertainty of the price drivers (covariance matrix defined by Eq. (5)). This input uncertainty is quantitatively described in Tables B.1 to B.4 in Appendix B, which reports the 5th, 25th, 50th, 75th, and 95th percentile values

of the corresponding uncertainty distributions. The Monte Carlo simulation is run for 500 price driver scenarios, with no significant variation being observed in the results for a higher number of scenarios.

The expected distribution of mean annual electricity prices are shown in Fig. 8-a, where the boxes indicate the interquartile range (i.e. 25th and 75th percentile) and the whiskers indicate the 5th and 95th percentile. Several price scenarios provided by the BEIS market-based model are reported for comparison. Due to the characterization of the uncertainty associated with the price drivers, (i) the uncertainty bounds of the price predictions increase with the forecast horizon, and (ii) the median of the price distribution for all years is consistent with the point forecast of the data-driven model in Fig. 7-b.

Most BEIS price scenarios fall within the interquartile range, suggesting that the data-driven model encompasses the market mechanisms underlying the market-based model. Only scenario “High prices” falls outside the interquartile range at the beginning of the time horizon and confirm the predictions provided by the proposed model. The median of the price distribution mostly correlates to the “Low prices” scenario, which best describes the median evolution of the price drivers.

The uncertainty distributions are positively skewed resulting in more scenarios of high electricity price, with upper bounds being around 140 EUR/MWh for 2035. In contrast, lower uncertainty bounds only decrease slightly below 40 EUR/MWh. Overall, a trend towards higher electricity prices is observed, which is mostly driven by an increase in natural gas price and solar generation, with most scenarios being characterized by a moderate increase.

Fig. 8-b shows hourly resolved price predictions for five exemplary days in 2020 (gray) and 2035 (light blue) with the identical residual profile superimposed on predicted price features. Two trends can be identified: (i) the annual mean price is expected to increase slightly (i.e. higher average value going from 2020 to 2035), and (ii) intra-day price volatility is increasing over time, resulting in more pronounced price peaks (i.e. higher difference between minimum and maximum values from 2020 to 2035).

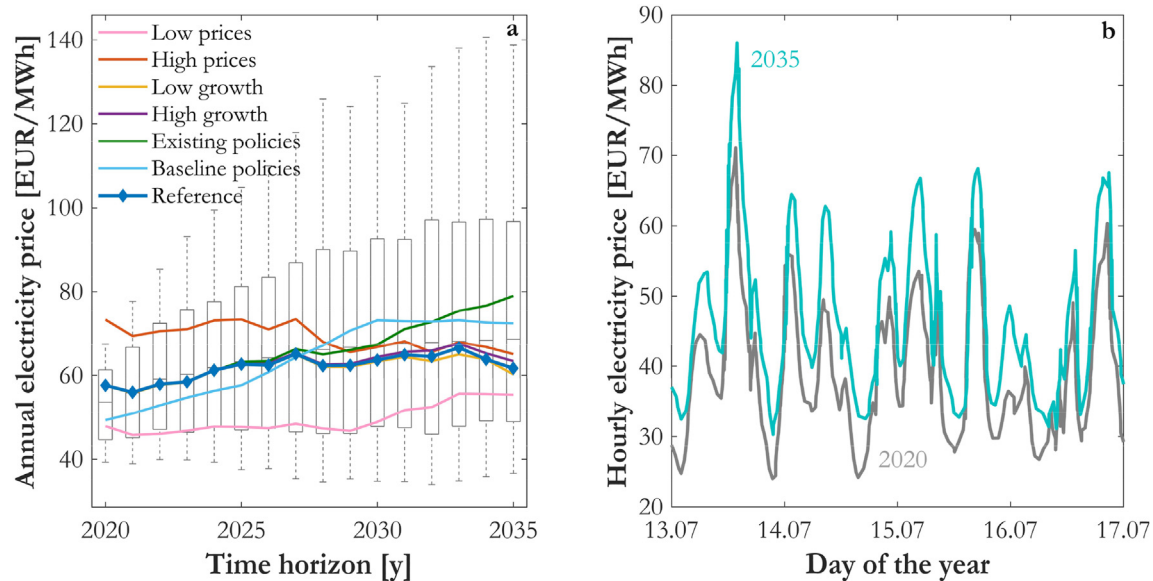


Fig. 8. (a) Distribution of electricity prices (box plot) resulting from stochastic price predictions with the box indicating the 25th and 75th percentile and the whiskers indicating the 5th and 95th percentile. Several BEIS price scenarios are also shown (coloured lines). (b) Five exemplary days of hourly resolved price predictions for 2020 (gray) and 2035 (cyan) with the identical residual profile superimposed on the predicted price features.

4. Conclusions

This paper addresses the challenges arising when predicting future electricity market prices over decade-long time horizons via data-driven models. We focus on two major challenges, namely (i) combining long time horizons with fine time resolutions, and (ii) predicting out-of-sample values of electricity price when considering future energy and market scenarios that have not been observed in the past. Remarkably, such data-driven model is able to not only predict annual or monthly electricity prices, but also generate generic, hourly-resolved future price profiles. Moreover, the data-driven approach is transferable across countries in two ways: (1) by coupling price drivers with market-based models to increase prediction accuracy and reduce prediction uncertainties; (2) by training the model on countries with fully available sets of price drivers and market-based model predictions, and generate predictions for countries with limited data sources available (e.g., without market-based predictions). Compared to market-based models, the data-driven approach relies solely on price driver data both for training and prediction, without the need for expert knowledge of market mechanisms specific to a country.

The prediction of hourly-resolved electricity price profiles over long time horizon is based on Fourier analysis. The electricity price profile is described through the amplitudes of the most relevant frequencies of the corresponding Fourier series, which represent a set of annual coefficients describing the main dynamics of the electricity price. A set of six frequencies is found to maximize the model predictive performance. These are the average, annual, weekly, daily, twelve- and eight-hourly frequencies. The corresponding coefficients of the price approximation are predicted through a regression model, which uses as input data annual values of relevant energy and market quantities, i.e. the price drivers. Different regression models are assessed and compared, namely linear regression, GPR and ANN. The model selection process shows that single-output GPR leads to the highest predictive performance.

When the model is trained and validated for a single country (UK in this case), hourly price approximations are predicted with a MAPE ranging from 7.6% to 19.3%, whereas errors on annual prices range from 3.6% to 16%. When making cross-country predictions, the model

is still able to capture the most relevant price dynamics, though a larger error of 37.1% is observed on 2015 mean annual prices due to discrepancies in absolute value of electricity prices among countries.

Whereas out-of-sample issues can arise even on long-term time horizons, these can be addressed by coupling data-driven and market-based models. When comparing data-driven and market-based models over the time horizon 2015–2019, comparable performance is observed.

Finally, we characterize the uncertainty associated with price predictions based on the uncertainty of price drivers. To this aim, the model is applied to determine future trends of the UK electricity market price while accounting for the uncertainty associated with the price drivers. The stochastic assessment shows that prices are most expected to moderately increase over the forecast horizon, that the intra-year variability is expected to increase, and that data-driven and market-based scenarios are in good agreement. Remarkably, most of the market-based price scenarios fall within the interquartile range of the data-driven prediction, suggesting the data-driven model can capture the market mechanisms underlying the market-based model.

Overall, findings support the relevance and validity of data-driven long-term predictions and highlight the potential for hybrid models where data-driven and market-based approaches are coupled. More specifically, hybrid models allow to exploit the ability of data-driven models to capture relevant price dynamics on time scales ranging from hourly to yearly, as well as the ability of market-based models to potentially anticipate and account for shifts in pricing regime or variations in the relationship between the price drivers and the electricity price.

Credit author statement

Conceptualization: PG; Formal analysis: MW, PG; Investigation: MW, PG; Methodology: all; Software: MW; Visualization: all; Writing – original draft: PG, MW; Writing – review & editing: all; Funding acquisition: PG, GS.

Declaration of competing interest

The authors declare that they have no known competing

financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Effect of data-resolution on correlations of electricity price and price drivers

Electricity prices exhibit strong hourly characteristics impacting the profitability of non-dispatchable generation assets.

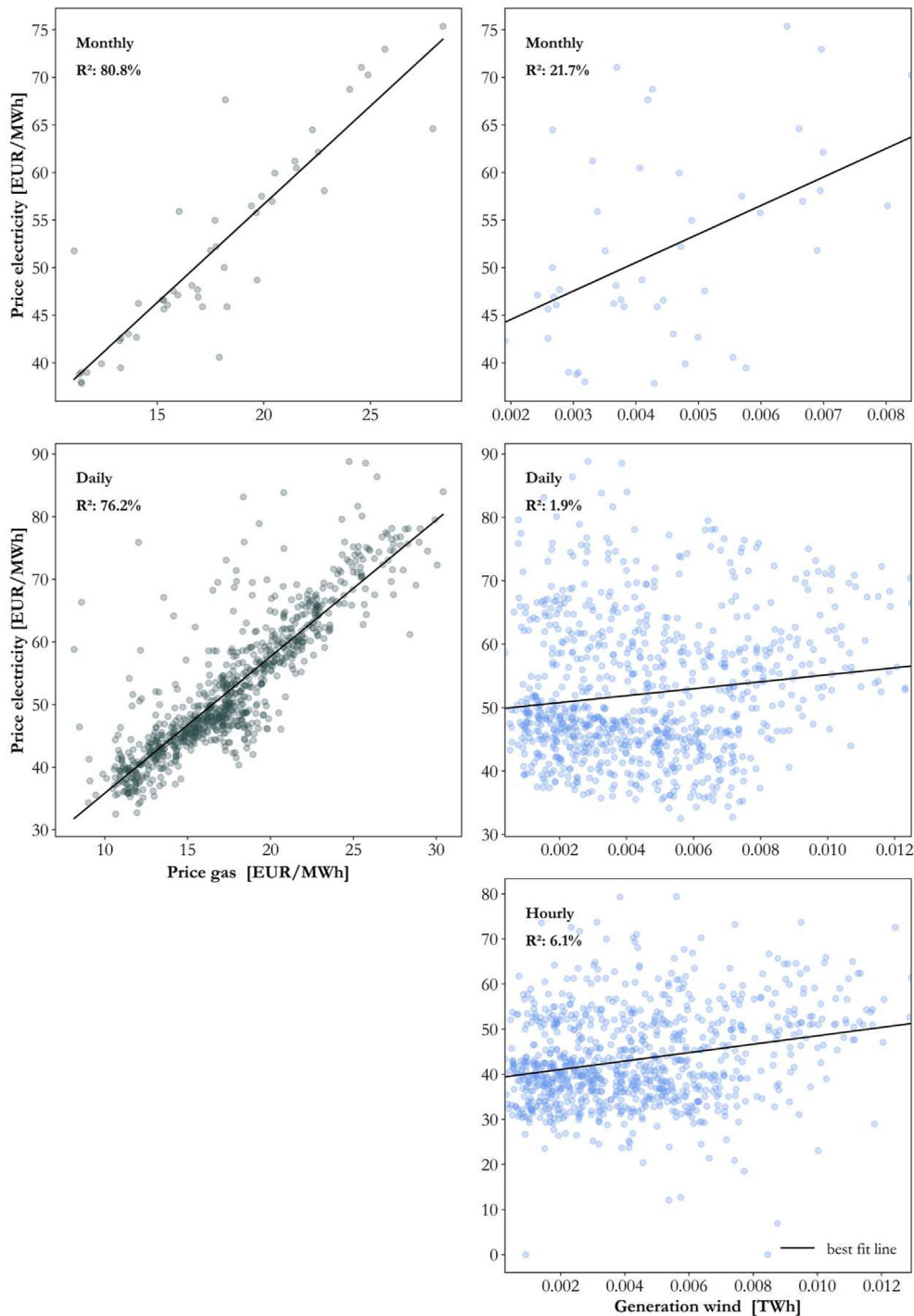


Fig. A.1. Electricity prices plotted against gas prices and wind generation, in monthly, daily and hourly resolution from 2015 to 2019 in the UK. The R^2 of the best fit line indicates a trend of stronger linear correlation with decreasing time resolution and very limited predictive value of the key drivers when analysing highly-resolved electricity prices.

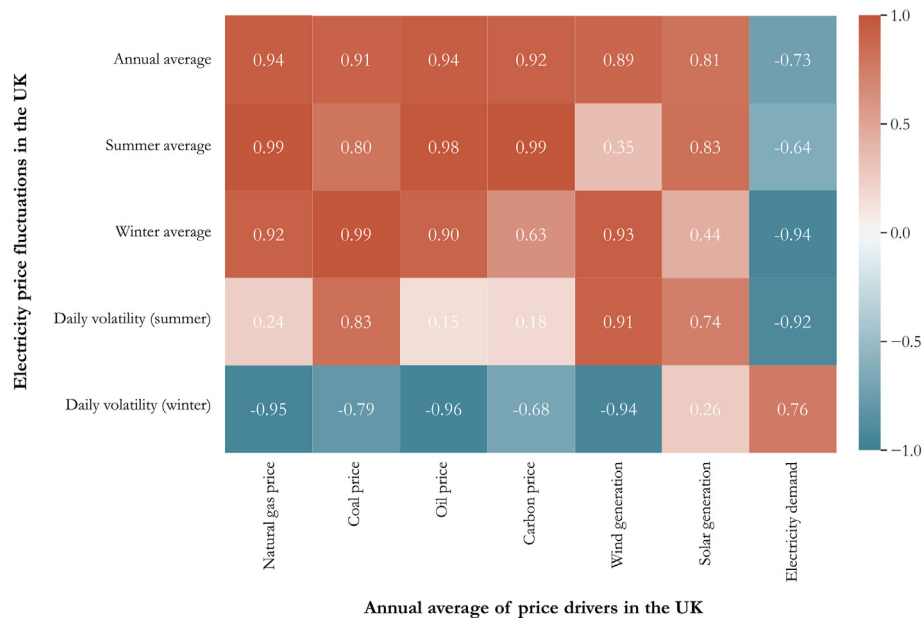


Fig. A.2. Pearson correlation coefficients of mean electricity prices (y-axis) varying from annual to hourly resolution as function of price drivers with the same resolution (x-axis), from 2015 to 2019 in the UK.

However, hourly forecasts of price drivers are not available when considering long-term prediction horizons. Furthermore, even with price drivers available in high resolution (e.g. daily resolution for fossil fuel prices and hourly resolution for renewable energy generation), essentially no correlation can be identified in most cases with reference to highly-resolved values of the electricity price. This is illustrated in Fig. A.1, which shows the electricity price as a function of two exemplary price drivers, namely gas price (left) and wind generation (right) from 2015 to 2019 in the UK. While there is a clear relationship between monthly gas and electricity prices, this relationship is much less significant on an daily level, and no hourly values are available for the natural gas price. Similarly, hourly wind generation does not show correlation with hourly electricity prices.

In contrast, a correlation exists between the base evolution of the electricity price and the average annual values of the price drivers, as illustrated in Fig. A.2, which reports the Pearson coefficients describing the correlation between the average annual values of some price drivers and different fluctuations of the electricity price for the UK, for the 2015–2019 time horizon.

Appendix B. Uncertainty characterization of input data

The uncertainty distributions associated with the model input data are characterized via an empirical prediction interval approach [49], using the reference scenario as point forecast (see Section 2.1). The stochastic assessment uses the following price drivers: solar generation (Table B.1), natural gas generation (Table B.2), natural gas price (Table B.3), and oil price (Table B.4). Their uncertainty is described through the 5th, 25th, 50th, 75th, and 95th percentile values of the corresponding uncertainty distributions.

Table B.1

Solar generation (MWh).

year	P5	P25	P50	P75	P95
2020	9.68	10.82	12.01	13.68	20.57
2021	7.90	10.19	11.99	14.20	26.39
2022	6.90	9.62	12.00	14.73	31.15
2023	6.08	9.29	12.00	15.31	35.03
2024	5.41	8.96	12.00	15.89	38.91
2025	4.76	8.83	12.00	16.36	42.68
2026	4.33	8.70	11.99	16.75	46.36
2027	3.93	8.59	11.99	17.29	49.94
2028	3.56	8.31	12.00	17.75	53.42
2029	3.08	7.87	12.04	18.01	56.85
2030	2.82	7.49	12.06	18.33	60.32
2031	2.43	7.16	12.05	18.95	63.78
2032	1.86	6.96	12.05	20.01	67.27
2033	1.28	7.05	12.05	20.99	70.76
2034	0.74	6.92	12.05	21.40	73.92
2035	0.84	7.00	12.06	21.93	76.47

Table B.2

Natural gas generation (MWh).

year	P5	P25	P50	P75	P95
2020	89.76	101.42	108.11	118.56	165.45
2021	91.42	103.74	111.57	123.46	182.32
2022	91.42	104.85	113.98	127.56	198.73
2023	89.04	106.08	116.23	131.30	208.87
2024	88.38	106.97	119.47	137.21	225.76
2025	91.01	108.32	122.61	139.59	234.59
2026	89.08	105.71	118.80	136.77	232.06
2027	85.42	102.60	115.45	132.93	228.41
2028	82.53	99.05	111.14	127.99	223.14
2029	77.80	94.98	107.19	125.69	219.52
2030	75.55	92.50	103.34	120.87	217.81
2031	80.12	97.93	107.66	126.22	228.40
2032	84.11	101.32	111.89	130.80	240.04
2033	87.85	105.08	116.27	135.82	250.22
2034	90.79	109.10	121.15	141.24	265.91
2035	94.58	113.78	126.03	147.06	276.11

Table B.3

Natural gas price (EUR/MWh).

year	P5	P25	P50	P75	P95
2020	30.05	30.37	35.09	35.23	36.67
2021	25.69	28.83	35.42	35.73	41.04
2022	24.48	30.06	36.78	39.18	49.02
2023	22.95	30.35	37.78	42.59	57.95
2024	21.91	31.64	39.64	46.32	67.88
2025	20.51	32.62	41.55	49.73	77.47
2026	18.13	31.88	42.39	51.89	86.77
2027	16.92	32.31	44.41	55.22	97.67
2028	14.76	31.80	45.50	57.61	108.48
2029	14.39	32.64	47.08	61.47	121.75
2030	13.64	33.13	48.22	65.88	134.58
2031	11.62	32.46	48.19	69.13	149.70
2032	9.16	30.89	48.18	72.23	165.03
2033	7.35	29.68	48.18	75.18	185.48
2034	5.97	28.85	48.19	78.52	207.13
2035	6.34	27.98	48.19	81.86	232.09

Table B.4

Oil price (EUR/MWh).

year	P5	P25	P50	P75	P95
2020	43.49	45.58	50.71	50.76	56.58
2021	38.49	41.38	50.11	50.15	59.85
2022	35.02	41.01	50.11	52.90	64.15
2023	32.99	41.87	50.90	58.20	71.76
2024	30.27	42.87	51.55	60.47	79.88
2025	27.77	44.08	52.42	62.25	89.25
2026	24.71	43.96	53.60	64.64	99.44
2027	22.99	42.47	54.14	66.22	108.37
2028	20.63	41.13	54.85	68.80	117.95
2029	18.82	40.22	56.14	72.41	128.65
2030	16.35	39.06	56.83	75.82	140.48
2031	13.57	37.95	57.01	79.40	152.77
2032	10.92	37.10	57.12	84.00	170.35
2033	9.68	36.67	57.16	86.92	188.48
2034	9.44	36.18	57.16	89.62	208.95
2035	7.64	35.46	57.17	92.65	230.34

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